

Dr Seamus Murphy
TÜV SÜD, 2187 Comox Ave,
Comox, British Columbia V9M 1P5, Canada
seamusrobertmurphy@gmail.com
ORCID 0000-0002-1792-0351

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Dr Christopher Ranger
Editor-in-Chief
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Royal Entomological Society

RE: Resubmission of AFE(2026)5804 as a new manuscript

Dear Dr Ranger,

Thank you for your decision of 22 September 2026 on manuscript AFE(2026)5804, "Testing the wind disruption and disturbance refugia hypothesis for mountain pine beetle outbreaks (*Dendroctonus ponderosae*)", and for inviting me to resubmit it as a new manuscript. You asked for an Introduction and Discussion better grounded in the existing literature and for an explicit statement of the study's novelty. Dr MacQuarrie asked that the Discussion link the findings to the biology, ecology and management of the beetle rather than repeat the Results. The attached "Author Response" quotes each of these comments and points to the section of the revised manuscript where it is addressed. In brief, additional literature was assembled under a written protocol that screened 1,257 unique records, which expanded the reference list from 30 to 71 works. The Introduction is redrafted on that literature, and the Discussion redrafted around the debates regarding dispersal, host defense, crown fade, landing and management.

The analysis was also updated during the revision to align with existing research on this specific method. Attack is now mapped in 47 sixteen-day Landsat periods with a classifier tested on field plots held out whole. Specifically, the models follow the order of Aukema et al. (2008), so that attack in neighbouring cells and in earlier periods is accounted for before any environmental variable is tested, and the results are checked under a latent spatial field and at four scales. In the revised analysis, the refugia presented in stands with smaller host trees. The wind interaction took the direction that plume disruption predicts but was too weak to establish it, and open ground had more attack, which is the pattern expected where wind-borne beetles settle. The author response lists every change of this kind.

The novelty of the study lies in the data that it tested. It entered a terrain-resolved wind field at the same sixteen-day interval as the attack maps, together with stand density and terms for attack in neighbouring cells and earlier periods. This design separated two predictions, a main effect of terrain shelter, expected if beetles landed where sheltered slopes slowed the wind, and an interaction between host density and wind, expected if wind broke up the pheromone plume that draws beetles to a tree. It also tested the three refugia mechanisms of Krawchuk et al. (2020) alongside the spread of the outbreak itself.

This study developed from its related study of forest regeneration on the same ground (Murphy et al. 2026, Forest Ecology and Management 618, 123985), whose field plots of beetle-killed pine provided the ground truth for the attack maps. The co-authors of that study are acknowledged on the title page. The work is original, unpublished and not under consideration elsewhere, and it received no external funding. I am the sole author and declare no conflict of interest. No one disputes ownership of the data, and every contribution is credited through authorship and acknowledgement. The derived data and the full analysis code are public on GitHub.

Yours sincerely,

Dr Seamus Murphy